

120. Physical Medium Attachment (PMA) sublayer, type 200GBASE-R and 400GBASE-R

120.1 Overview

120.1.1 Scope

This clause specifies the Physical Medium Attachment sublayer (PMA) for the families of 200 Gb/s and 400 Gb/s Physical Layer implementations, known as 200GBASE-R and 400GBASE-R. The PMA allows the PCS (specified in Clause 119) to connect in a media-independent way with a range of physical media, or for the DTE XS to connect to the PHY XS (specified in Clause 118). The 200GBASE-R PMA(s) can support any of the 200 Gb/s PMDs in Table 116–1, and the 400GBASE-R PMA(s) can support any of the 400 Gb/s PMDs in Table 116–2. The terms 200GBASE-R and 400GBASE-R are used when referring generally to Physical Layers using the PMA defined in this clause.

200GBASE-R and 400GBASE-R can be extended to support any full duplex medium requiring only that the PMD be compliant with the appropriate PMA interface.

The interfaces for the inputs of the 200GBASE-R and 400GBASE-R PMAs are defined in an abstract manner and do not imply any particular implementation. For 200GBASE-R PMAs, electrical interfaces connecting PMA sublayers, known as 200GAUI-n, are defined in Annex 120B, Annex 120C, Annex 120D, and Annex 120E. For 400GBASE-R PMAs, electrical interfaces connecting PMA sublayers, known as 400GAUI-n, are defined in Annex 120B, Annex 120C, Annex 120D, and Annex 120E.

120.1.2 Position of the PMA in the 200GBASE-R and 400GBASE-R sublayers

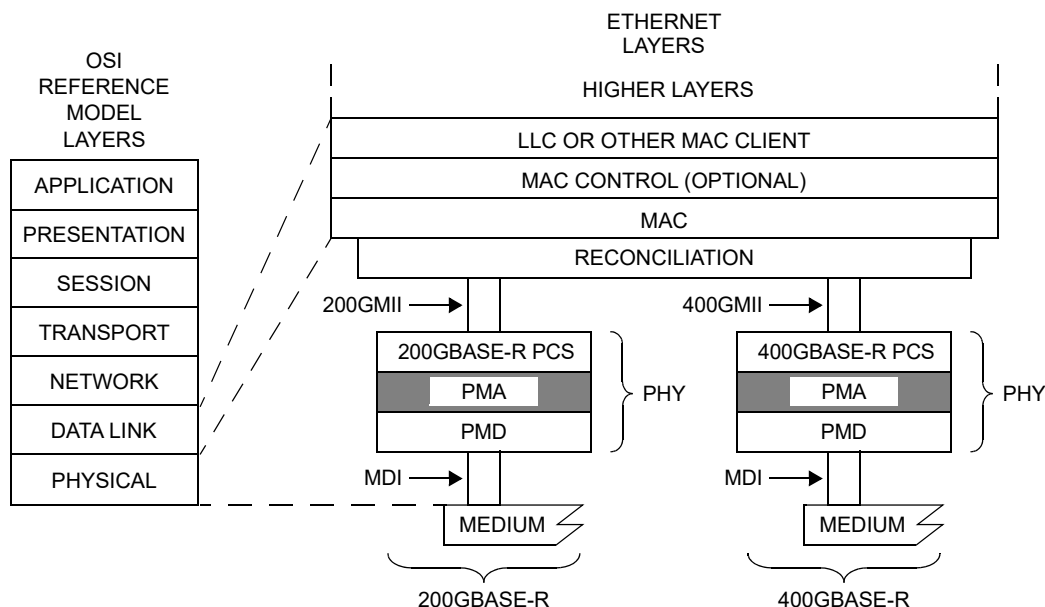
Figure 120–1 shows the relationship of the PMA sublayer (shown shaded) with other sublayers to the ISO Open System Interconnection (OSI) reference model.

120.1.3 Summary of functions

The following is a summary of the principal functions implemented (when required) by the PMA in both the transmit and receive directions:

- a) Adapt the PCSL formatted signal to the appropriate number of abstract or physical lanes.
- b) Provide per input-lane clock and data recovery.
- c) Provide bit-level multiplexing.
- d) Provide clock generation.
- e) Provide signal drivers.
- f) Optionally provide local loopback to/from the PMA service interface.
- g) Optionally provide remote loopback to/from the PMD service interface.
- h) Optionally provide test-pattern generation and detection.
- i) Tolerate Skew Variation.
- j) Perform PAM4 encoding and decoding for 200GBASE-R PMAs where the number of physical lanes is 4, and for 400GBASE-R PMAs where the number of physical lanes is 4 or 8.

In addition, the PMA provides receive link status information in the receive direction.



200GMII = 400 Gb/s MEDIA INDEPENDENT INTERFACE
400GMII = 400 Gb/s MEDIA INDEPENDENT INTERFACE
LLC = LOGICAL LINK CONTROL
MAC = MEDIA ACCESS CONTROL
MDI = MEDIUM DEPENDENT INTERFACE

PCS = PHYSICAL CODING SUBLAYER
PHY = PHYSICAL LAYER DEVICE
PMA = PHYSICAL MEDIUM ATTACHMENT
PMD = PHYSICAL MEDIUM DEPENDENT

Figure 120–1—200GBASE-R and 400GBASE-R PMA relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and IEEE 802.3 Ethernet model

120.1.4 PMA sublayer positioning

An implementation may use one or more PMA sublayers to adapt the number and rate of the PCS lanes to the number and rate of the PMD lanes. The number of PMA sublayers required depends on the partitioning of functionality for a particular implementation. An example is illustrated in Figure 120–2. Additional examples are illustrated in Annex 120A. Each PMA maps the PCSs from p PMA input lanes to q PMA output lanes in the Tx direction, and from q PMA input lanes to p PMA output lanes in the Rx direction.

Management Data Input/Output (MDIO) Manageable Device (MMD) addresses 1, 8, 9, and 10 are available for addressing multiple instances of PMA sublayers (see Table 45–1 for MMD device addresses). If the PMA sublayer that is closest to the PMD is packaged with the PMD, it shares MMD 1 with the PMD. More addressable instances of PMA sublayers, each one separated from lower addressable instances by chip-to-chip interfaces, may be implemented and addressed allocating MMD addresses to PMAs in increasing numerical order going from the PMD toward the MAC. The example shown in Figure 120–2 could be implemented with three addressable instances: MMD 1 addressing the lowest PMA sublayer (co-packaged with the PMD), MMD 8 addressing the PMA sublayer above the 200GAUI-4 below the 200GAUI-8 or above the 400GAUI-8 below the 400GAUI-16, and MMD 9 addressing the PMA sublayer below the PCS.

The number of input lanes and the number of output lanes for a PMA are always divisors of the number of PCSs. For PMA sublayers supporting 200GBASE-R PMDs, the number of PCSs is 8. For PMA sublayers supporting 400GBASE-R PMDs, the number of PCSs is 16.

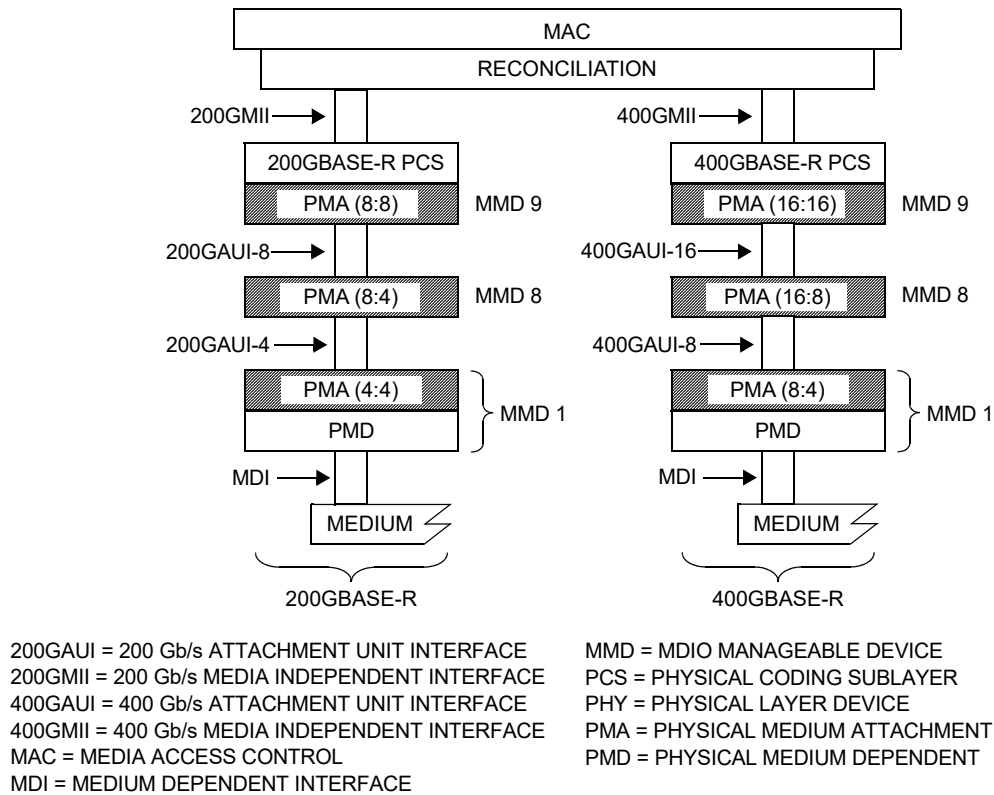


Figure 120-2—Example 200GBASE-R and 400GBASE-R PMA layering

The following guidelines apply to the partitioning of PMAs:

- a) The inter-sublayer service interface, defined in 116.3.1, is used for the PMA service interfaces supporting a flexible architecture with multiple PMA sublayers.
 - 1) An instance of this interface can only connect service interfaces with the same number of lanes, where the lanes operate at the same rate.
- b) 200GAUI-n is a physical instantiation of the connection between two adjacent 200GBASE-R PMA sublayers with the exception of the inst:IS_SIGNAL.indication which is carried outside of this physically instantiated interface. 400GAUI-n is a physical instantiation of the connection between two adjacent 400GBASE-R PMA sublayers with the exception of the inst:IS_SIGNAL.indication which is carried outside of this physically instantiated interface.
 - 1) As physical instantiations, these define electrical and timing specification as well as requiring a receive re-timing function.
 - 2) 200GAUI-8 is a 26.5625 GBd by 8 lane NRZ physical instantiation of the 200 Gb/s connection. 400GAUI-16 is a 26.5625 GBd by 16 lane NRZ physical instantiation of the 400 Gb/s connection.
 - 3) 200GAUI-4 is a 26.5625 GBd by 4 lane PAM4 physical instantiation of the 200 Gb/s connection. 400GAUI-8 is a 26.5625 GBd by 8 lane PAM4 physical instantiation of the 400 Gb/s connection.
- c) The abstract inter-sublayer service interface can be physically instantiated as a 200GAUI-n or 400GAUI-n, using associated PMAs to map to the appropriate number of lanes.
- d) Opportunities for optional test-pattern generation, optional test-pattern detection, optional local loopback and optional remote loopback are dependent upon the location of the PMA sublayer in the implementation. See Figure 120-5.

- e) A minimum of one PMA sublayer is required in a PHY.
- f) A maximum of four PMA sublayers are addressable as MDIO MMDs.

120.2 PMA interfaces

All PMA variants for 200GBASE-R and 400GBASE-R are based on a generic specification of a bit mux function that applies to all input/output lane counts and each direction of transmission. Each direction of transmission may employ one or more such bit muxes to adapt from the appropriate number of input lanes to the appropriate number of output lanes as illustrated in Figure 120–3.

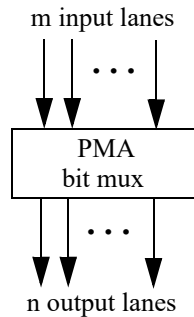


Figure 120–3—PMA bit mux used in both Tx and Rx directions

Conceptually, the PMA bit mux operates in one direction of transmission by demultiplexing PCSs from m PMA input lanes and remultiplexing them into n PMA output lanes. The mapping of PCSs from input to output lanes is not specified. See 120.5.2 and Figure 120–4 for details.

Figure 120–5 provides the functional block diagram of a PMA. The parameters of a PMA include the following:

- The numbers of input and output lanes in each direction.
- Whether the PMA is adjacent to a physically instantiated interface (200GAUI- n or 400GAUI- n above or below).
- Whether the PMA is adjacent to the PCS or DTE XS.
- Whether the PMA is adjacent to the PMD or PHY XS.

120.3 PMA service interface

The PMA service interface for 200GBASE-R and 400GBASE-R is an instance of the inter-sublayer service interface defined in 116.3.1. The PMA service interface primitives are summarized as follows:

```
PMA:IS_UNITDATA_ i.request(tx_symbol)
PMA:IS_UNITDATA_ i.indication(rx_symbol)
PMA:IS_SIGNAL.indication(SIGNAL_OK)
```

For a PMA service interface with p lanes, the primitives are defined for $i = 0$ to $p - 1$.

If the PMA client is the PCS or DTE XS, the PMA continuously sends eight (for 200GBASE-R) or sixteen (for 400GBASE-R) parallel bit streams to the PCS, each at the nominal signaling rate of the PCSL.

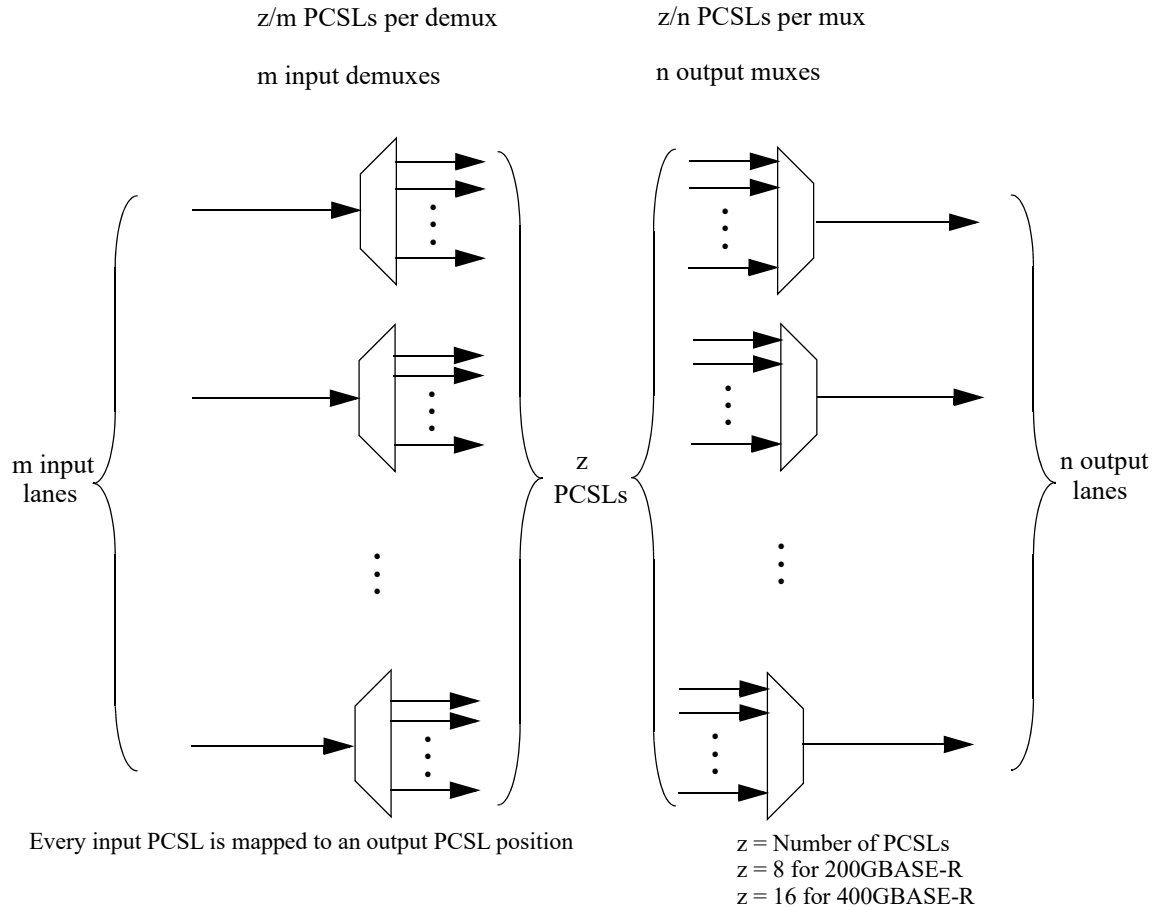


Figure 120–4—PMA bit mux operation used in both Tx and Rx directions

If a PMA client is another PMA, a 200GBASE-R PMA with $p=8$ or a 400GBASE-R PMA with $p=16$ physical input lanes receives bits from one PCSL on each of its input lanes. A 200GBASE-R PMA with $p=4$ physical input lanes receives PAM4 symbols on each of its input lanes at the PCSL rate, each symbol formed from two bits. A 400GBASE-R PMA with $p=4$ or $p=8$ physical input lanes receives PAM4 symbols on each of its input lanes at $8/p$ times the PCSL rate, each symbol formed from two bits. The bit stream represented by the input symbols carries z/p bit-multiplexed PCSLs on each physical input lane, where $z=8$ for 200GBASE-R and $z=16$ for 400GBASE-R. Skew may exist between different PCSLs received on the same physical input lane even though all PCSLs originate from the same synchronous source, so there is independence of arrival of bits from each PCSL on each physical input lane.

In the Tx direction, if the symbol from a PMA:IS_UNITDATA_ i .request primitive is received over a physically instantiated interface (200GAUI- n or 400GAUI- n), clock and data are recovered on the lane receiving the symbol. If necessary, PAM4 symbols are converted to pairs of bits. Bits are routed through the PMA to an output lane through a process that may demultiplex PCSLs from the input, perform any necessary buffering to tolerate Skew Variation across input lanes, and multiplex PCSLs to output lanes. If necessary, pairs of bits on output lanes are converted to PAM4 symbols. The symbols are sent on an output lane to the sublayer below using the *inst*:IS_UNITDATA_ k .request (k not necessarily equal to i) primitive (see 120.4).

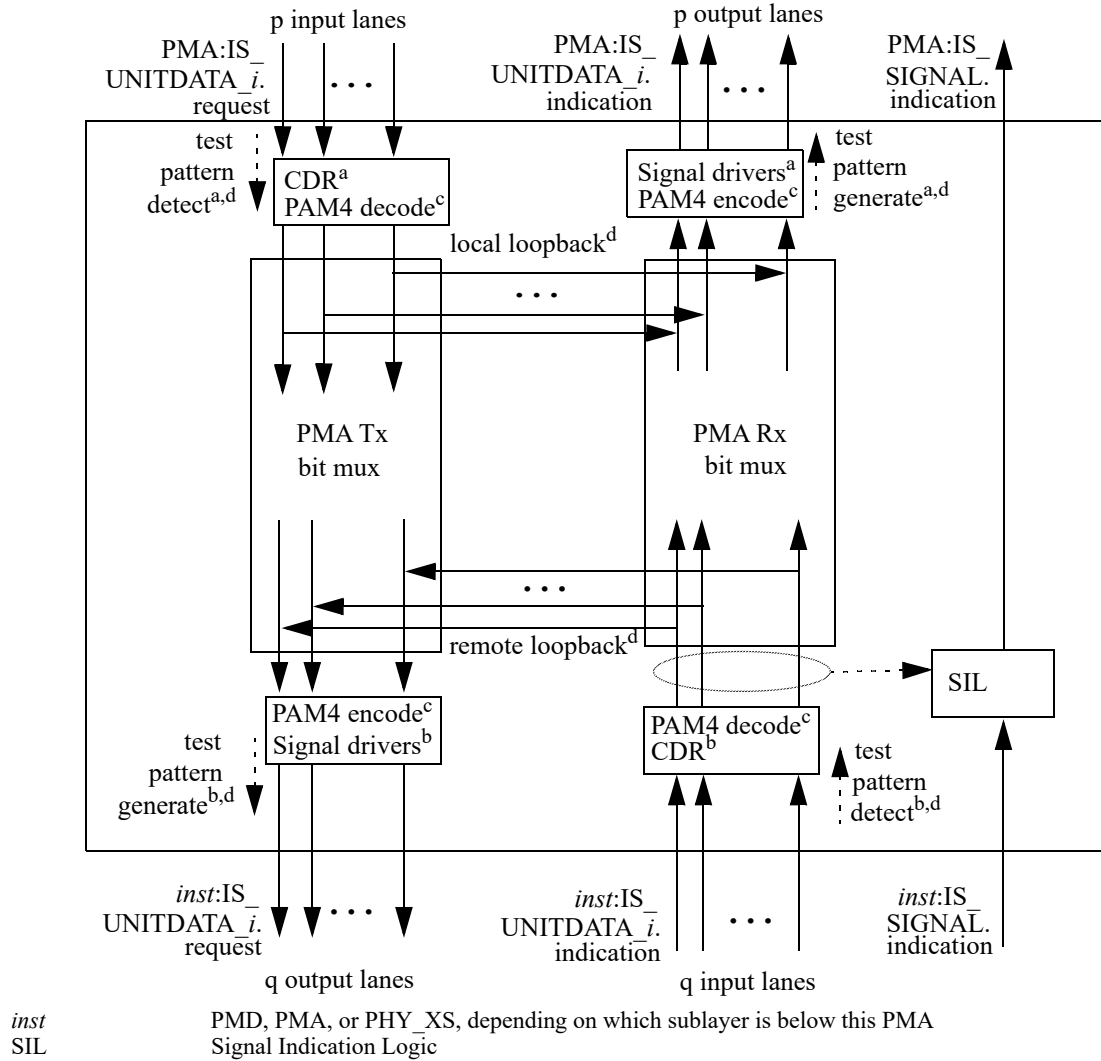


Figure 120-5—PMA Functional Block Diagram

The PMA passes symbols from the input lanes to the output lanes in the Rx direction when data is being received from every input lane from the sublayer below the PMA that has a PCSL that is routed to a particular output lane at the PMA service interface, and (if necessary), buffers are filled to allow tolerating the Skew Variation that may appear between the input lanes. PCSLs are demultiplexed from the input lanes, remultiplexed to the output lanes, and symbols are transferred over each output lane to the PMA client via the PMA:IS_UNITDATA_i .indication primitive. If necessary, PAM4 symbols are converted to pairs of bits on the input lanes and/or pairs of bits are converted to PAM4 symbols on the output lanes.

The PMA:IS_SIGNAL .indication primitive is generated through a set of Signal Indication Logic (SIL) that reports signal health based on receipt of the inst:IS_SIGNAL .indication from the sublayer below, data being received on all of the input lanes from the sublayer below, buffers filled (if necessary) to accommodate Skew Variation, and symbols being sent to the PMA client on all of the output lanes. When these conditions

are met, the SIGNAL_OK parameter sent to the PMA client via the PMA:IS_SIGNAL.indication primitive has the value OK. Otherwise, the SIGNAL_OK primitive has the value FAIL. In the case where the sublayer below the PMA is a PHY XS the PMA does not receive a PHY_XS:IS_SIGNAL.indication as an input to the SIL.

120.4 Service interface below PMA

Since the architecture supports multiple PMA sublayers for various PMD lane counts and device partitioning, there are several different sublayers that may appear below a PMA, including the PMD, an extender sublayer, or another PMA. The variable *inst* represents whichever sublayer appears below the PMA (e.g., another PMA or PMD).

The sublayer below the PMA utilizes the inter-sublayer service interface defined in 116.3. The service interface primitives provided to the PMA are summarized as follows:

```
inst:IS_UNITDATA_i.request(tx_symbol)
inst:IS_UNITDATA_i.indication(rx_symbol)
inst:IS_SIGNAL.indication(SIGNAL_OK)
```

The number of lanes *q* for the service interface matches the number of lanes expected by the PMA. The *inst:IS_UNITDATA_i* primitives are defined for each lane *i* = 0 to *q* – 1 of the service interface below the PMA. Note that electrical and timing specifications of the service interface are defined if the interface is physically instantiated (e.g., 200GAUI-n or 400GAUI-n), otherwise the service interface is specified only abstractly. The interface between the PMA and the sublayer below consists of *q* lanes for data transfer and a status indicating a good signal from the sublayer below the PMA (see Figure 120–5).

The PMA transfers symbols from the input lanes to the output lanes in the Tx direction when data is being received via the PMA:IS_UNITDATA_i.request primitive from every input lane from the PMA client at the PMA service interface (see 120.3) that has a PCSL that is routed to this output lane, and (if necessary), buffers are filled to provide the ability to tolerate the Skew Variation that may appear between the input lanes from the PMA client. PCSLs are demultiplexed from the input lanes, remultiplexed to the output lanes, and symbols are transferred over each output lane to the sublayer below the PMA.

In the Rx direction, if the symbol is received over a physically instantiated interface (200GAUI-n, 400GAUI-n, or physically instantiated PMD service interface), clock and data are recovered on the lane receiving the symbol. If necessary, PAM4 symbols received on the input lanes are converted to pairs of bits. The bits are routed through the PMA to an output lane toward the PMA client through a process that may demultiplex PCSLs from the input, perform any necessary buffering to tolerate Skew Variation across input lanes, and multiplex PCSLs to output lanes. If necessary, pairs of bits are converted to PAM4 symbols on the output lanes. Each symbol is sent on an output lane to the PMA client using the PMA:IS_UNITDATA_k.indication (*k* not necessarily equal to *i*) primitive at the PMA service interface.

In the case where the sublayer below the PMA is a PHY XS, there is an additional primitive:

```
PHY_XS:IS_SIGNAL.request(SIGNAL_OK)
```

The PHY_XS:IS_SIGNAL.request primitive is generated through a set of SIL that reports signal health based on data being received on all of the input lanes from the sublayer above, buffers filled (if necessary) to accommodate Skew Variation, and symbols being sent to the PHY XS on all of the output lanes. When these conditions are met, the SIGNAL_OK parameter sent to the PHY XS via the PHY_XS:IS_SIGNAL.request primitive has the value OK. Otherwise, the SIGNAL_OK primitive has the value FAIL.

120.5 Functions within the PMA

The purpose of the PMA is to adapt the PCSL formatted signal to an appropriate number of abstract or physical lanes, to recover clock from the received signal (if appropriate), and optionally to provide test signals and loopback. Each input (Tx direction) or output (Rx direction) lane between the PMA and the PMA client carries one or more PCSLs that are bit-multiplexed. All input and output lanes between the PMA and the PMA client carry the same number of PCSLs and operate at the same nominal signaling rate. Likewise, each input (Rx direction) or output (Tx direction) lane between the PMA and the sublayer below the PMA carries one or more PCSLs that are bit-multiplexed. All input and output lanes between the PMA and the sublayer below the PMA carry the same number of PCSLs and operate at the same nominal signaling rate. As described in 120.1.4, the number of input lanes and the number of output lanes for a given PMA are divisors of 8 for 200GBASE-R, or 16 for 400GBASE-R, which are the number of PCSLs for the respective PHYs. The symbols received from input lanes or sent on output lanes are bits for 200GBASE-R interfaces where the number of lanes is 8, or 400GBASE-R interfaces where the number of lanes is 16. The symbols received from input lanes or sent on output lanes are PAM4 symbols for 200GBASE-R interfaces where the number of lanes is 4, or 400GBASE-R interfaces where the number of lanes is 8 or 4.

120.5.1 Per input-lane clock and data recovery

If the interface between the PMA client and the PMA is physically instantiated as 200GAUI-n or 400GAUI-n, the PMA shall meet the electrical and timing specifications in Annex 120B, Annex 120C, Annex 120D, or Annex 120E as appropriate. If the interface between the sublayer below the PMA and the PMA is physically instantiated as 200GAUI-n or 400GAUI-n, the PMA shall meet the electrical and timing specifications at the service interface below the PMA as specified in Annex 120B, Annex 120C, Annex 120D, or Annex 120E as appropriate.

Test patterns that are intended for transmitter testing, such as square wave, may not be correctly recovered by an adjacent PMA.

120.5.2 Bit-level multiplexing

The PMA provides bit-level multiplexing in both the Tx and Rx directions. In the Tx direction, the function is performed among the bits received from the PMA client via the `PMA:IS_UNITDATA_i.request` primitives (for PMA client lanes $i = 0$ to $p - 1$) with the result sent to the service interface below the PMA using the `inst:IS_UNITDATA_i.request` primitives (for service interface lanes $i = 0$ to $q - 1$), referencing the functional block diagram shown in Figure 120–5. The bit multiplexing behavior is illustrated in Figure 120–4.

The aggregate signal carried by the group of input lanes or the group of output lanes is arranged as a set of PCSLs. The number of PCSLs z is 8 for 200GBASE-R interfaces and 16 for 400GBASE-R interfaces. The nominal bit rate R of each PCSL is 26.5625 Gb/s.

For a PMA with m input lanes (Tx or Rx direction), each input lane carries, bit multiplexed, z/m PCSLs. Each input lane has a nominal bit rate of $26.5625 \times z/m$ Gb/s. Note that the signaling (Baud) rate is equal to the bit rate when the number of physical lanes is 8 for 200GBASE-R or 16 for 400GBASE-R (bits are sent or received on the lanes). The Baud rate is equal to half of the bit rate when the number of physical lanes is 4 for 200GBASE-R or the number of physical lanes is 8 or 4 for 400GBASE-R (PAM4 symbols are sent or received on the lanes). If necessary, PAM4 symbols are converted to pairs of bits on the input lanes and/or pairs of bits are converted to PAM4 symbols on the output lanes. If bit x received on an input lane belongs to a particular PCSL, the next bit of that same PCSL is received on the same input lane at bit position $x + (z/m)$. The z/m PCSLs may arrive in any sequence on a given input lane.

For a PMA with n output lanes (Tx or Rx direction), each output lane carries, bit multiplexed, z/n PCSLs. Each output lane has a nominal signaling rate of $26.5625 \times z/n$ Gb/s. Each PCSL is mapped from a position

in the sequence on one of the m input lanes to a position in the sequence on one of the n output lanes. If bit x sent on an output lane belongs to a particular PCSL, the next bit of that same PCSL is sent on the same output lane at bit position $x + (z/n)$. The PMA shall maintain the chosen sequence of PCSLs on all output lanes while it is receiving a valid stream of bits on all input lanes.

Each PCSL received in any temporal position on an input lane is transferred into a temporal position on an output lane. As the PCS (see Clause 119) has fully flexible receive logic, an implementation is free to perform the mapping of PCSLs from input lanes to output lanes without constraint. Figure 120–6 illustrates one possible bit ordering for a 400GBASE-R 8:4 PMA bit mux. Other bit orderings are also valid.

Note that since the number of input lanes and output lanes for a 200GBASE-R or 400GBASE-R PMA is always a power of two, many PMAs converting between different numbers of lanes normally simply multiplex two or four input lanes to one output lane, or demultiplex two or four output lanes from one input lane. However, any PMA implementation which produces an allowable order of bits from all PCSLs on the output lanes is valid.

NOTE—PMA output lanes composed of some specific combinations of four PCSLs with specific skew offsets (e.g., 400GBASE-R PCSLs 0, 2, 4, and 10 with delays 0, 1, 0, and 2 bits, respectively) may have reduced transition density.

120.5.3 Skew and Skew Variation

The Skew (relative delay) between the PCSLs is kept within limits so that the information on the lanes can be reassembled by the PCS.

Any PMA that combines PCSLs from different input lanes onto the same output lane is required to tolerate Skew Variation between the input lanes without changing the PCSL positions on the output. Skew and Skew Variation are defined in 116.5. The limits for Skew and Skew Variation at physically instantiated interfaces are specified at Skew points SP1 and SP2 in the transmit direction and SP5 and SP6 in the receive direction as defined in 116.5 and illustrated in Figure 116–4 and Figure 116–5.

120.5.3.1 Skew generation toward SP1

In an implementation with one or more physically instantiated 200GAUI-n or 400GAUI-n interfaces, the PMA that sends data in the transmit direction toward the 200GAUI-n or 400GAUI-n that is closest to the PMD (SP1 in Figure 116–4 and Figure 116–5) shall produce no more than 29 ns of Skew between PCSLs toward the 200GAUI-n or 400GAUI-n, and no more than 200 ps of Skew Variation.

120.5.3.2 Skew tolerance at SP1

In an implementation with one or more physically instantiated 200GAUI-n or 400GAUI-n interfaces, the PMA service interface that receives data in the transmit direction from the 200GAUI-n or 400GAUI-n (SP1 in Figure 116–4 and Figure 116–5) shall tolerate the maximum amount of Skew Variation allowed at SP1 (200 ps) between input lanes while maintaining the bit ordering and position of each PCSL on each PMA lane in the transmit direction (toward the PMD).

120.5.3.3 Skew generation toward SP2

If the PMD service interface is physically instantiated so that the Skew at SP2 can be measured, the PMA adjacent to the PMD service interface shall generate no more than 43 ns of Skew, and no more than 400 ps of Skew Variation between output lanes toward the PMD service interface (SP2 in Figure 116–4 and Figure 116–5). If there is a physically instantiated 200GAUI-n or 400GAUI-n as well, then the Skew measured at SP1 is limited to no more than 29 ns of Skew and no more than 200 ps of Skew Variation.

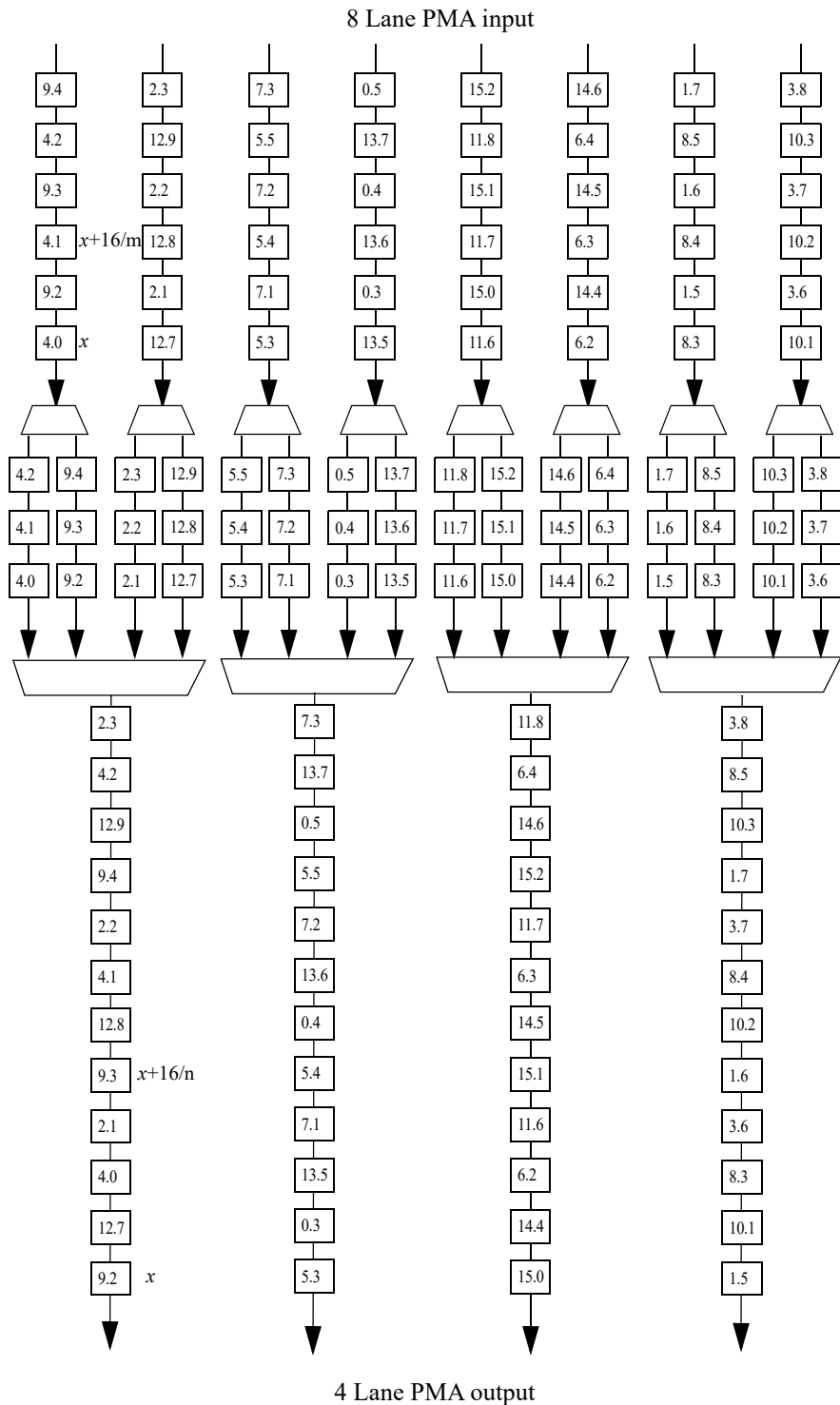


Figure 120–6—Example 8:4 400GBASE-R PMA bit mux

120.5.3.4 Skew tolerance at SP5

If the PMD service interface is physically instantiated so that the Skew at SP5 can be measured, the PMA adjacent to the PMD service interface (SP5) shall tolerate the maximum amount of Skew Variation allowed at SP5 (3.6 ns) between output lanes from the PMD service interface while maintaining the bit ordering and position of each PCSL on each PMA lane in the receive direction (toward the MAC).

120.5.3.5 Skew generation at SP6

In an implementation with one or more physically instantiated 200GAUI-n or 400GAUI-n interfaces, at SP6 (the receive direction of the 200GAUI-n or 400GAUI-n closest to the PCS), the PMA or group of PMAs between the PMD and the 200GAUI-n or 400GAUI-n closest to the PCS shall deliver no more than 160 ns of Skew, and no more than 3.8 ns of Skew Variation between output lanes toward the 200GAUI-n or 400GAUI-n in the Rx direction.

If there is a physically instantiated PMD service interface that allows the Skew to be measured, the Skew measured at SP5 is limited to no more than 145 ns of Skew and no more than 3.6 ns of Skew Variation. If there is no physically instantiated PMD service interface, the Skew measured at SP4 is limited to no more than 134 ns of Skew, and no more than 3.4 ns of Skew Variation.

120.5.3.6 Skew tolerance at SP6

In an implementation with one or more physically instantiated 200GAUI-n or 400GAUI-n interfaces, the PMA between the 200GAUI-n or 400GAUI-n closest to the PCS and the PCS shall tolerate the maximum amount of Skew Variation allowed at SP6 (3.8 ns) between input lanes while maintaining the bit order and position of PCSLs on lanes sent in the receive direction towards the MAC.

120.5.4 Delay constraints

The maximum cumulative delay contributed by up to four PMA stages in a PHY (sum of transmit and receive delays at one end of the link) shall meet the values specified in Table 120–1. A description of overall system delay constraints and the definitions for bit-times and pause_quanta can be found in 116.4 and its references.

Table 120–1—Delay constraints

Sublayer	Maximum (bit time)	Maximum (pause_quanta)	Maximum (ns)
200GBASE-R PMA	18 432	36	92.16
400GBASE-R PMA	36 864	72	92.16

120.5.5 Clocking architecture

A PMA with m input lanes and n output lanes shall clock the output lanes such that the bit rate of the output lanes is m/n times the bit rate of the input lanes. For input or output lanes encoded as PAM4 (for 200GBASE-R, where the number of input or output lanes is 4, or for 400GBASE-R, where the number of input or output lanes is 4 or 8), the symbol rate is half of the bit rate. This applies in both the Tx and Rx directions of transmission. In the case where the interfaces between the PMA client and the PMA and/or the PMA and the sublayer below the PMA are physically instantiated, the PMA may derive its input clock(s)

from the electrical interface on one or more of the input lanes, and generate the output clock(s) with an appropriate PLL multiplier/divider circuit.

NOTE—For a PMA where the signaling rate on each output lane is higher than the signaling rate on each input lane, any low frequency jitter on the input lanes may result in more jitter relative to the UI on the output lanes.

There is no requirement that the PMA clock all output lanes in unison. Examples of independent clocking of output lanes include the following:

- The case where the number of input and output lanes are equal (the PMA is provided for retiming and regeneration of the signal). This may be implemented without any rearrangement of PCSs between input lanes and output lanes (although rearrangements are allowed), and such a PMA may be implemented by driving each output lane using the clock recovered from the corresponding input lane.
- Since the number of PCSs is a power of 2, whenever the number of input and output lanes are unequal, each input lane could be mapped to multiple output lanes (in the case where the number of output lanes is greater), or multiple input lanes could be mapped to a single output lane.

120.5.6 Signal drivers

For cases where the interface between the PMA client and the PMA, or between the PMA and the sublayer below the PMA represent a physically instantiated interface, the PMA provides electrical signal drivers for that interface. The electrical and jitter/timing specifications for these interfaces appear in

- Annex 120B, which specifies the 200GAUI-8 and 400GAUI-16 interfaces for chip-to-chip applications.
- Annex 120C, which specifies the 200GAUI-8 and 400GAUI-16 interfaces for chip-to-module applications.
- Annex 120D, which specifies the 200GAUI-4 and 400GAUI-8 interfaces for chip-to-chip applications.
- Annex 120E, which specifies the 200GAUI-4 and 400GAUI-8 interfaces for chip-to-module applications.

For 200GAUI-8 or 400GAUI-16, the modulation format is NRZ. For 200GAUI-4 or 400GAUI-8, the modulation format is PAM4.

120.5.7 PAM4 Encoding

120.5.7.1 Gray mapping for PAM4 encoded lanes

For output lanes encoded as PAM4 (for 200GBASE-R, where the number of output lanes is 4, or for 400GBASE-R, where the number of output lanes is 4 or 8), the PMA transmit process shall map consecutive pairs of bits {A, B}, where A is the bit arriving first, to a Gray-coded symbol as follows:

{0, 0} maps to 0,
{0, 1} maps to 1,
{1, 1} maps to 2, and
{1, 0} maps to 3.

For input lanes encoded as PAM4 (for 200GBASE-R, where the number of input lanes is 4, or for 400GBASE-R, where the number of input lanes is 4 or 8), the PMA receive process shall map Gray-coded PAM4 symbols to pairs of bits {A, B} where A is considered to be the first bit as follows:

0 maps to {0, 0},
1 maps to {0, 1},

2 maps to {1, 1}, and
3 maps to {1, 0}.

120.5.7.2 Precoding for PAM4 encoded lanes

The precoding specifications in this subclause apply to the input and output lanes of a PMA that are connected to the service interface of a 200GBASE-R PMD that includes the PMD control function defined in 136.8.11 (200GBASE-CR4 or 200GBASE-KR4).

The PMA shall provide $1/(1+D) \bmod 4$ precoding capability on each transmit lane and may optionally provide $1/(1+D) \bmod 4$ decoding capability on each receive lane. Precoding is implemented as specified in 135.5.7.2.

The precoder is enabled independently on the Tx output (toward the PMD) and the Rx input (from the PMD) on each lane (0, 1, 2, and 3). Precoding is enabled and disabled using variables `precoder_tx_out_enablei` and `precoder_rx_in_enablei` (where *i* is 0, 1, 2, or 3).

If a Clause 45 MDIO is implemented, the variables `precoder_tx_out_enablei` and `precoder_rx_in_enablei` are accessible through register 1.600 and 1.601 (see 45.2.1.139 and 45.2.1.140).

If the PMA is connected to the service interface of a PMD that includes the PMD control function and training is enabled by the management variable `mr_training_enable` (see 136.7), then `precoder_tx_out_enablei` and `precoder_rx_in_enablei` shall be set as determined by the PMD control function in the LINK_READY state on lane *i* (see 136.8.11.7.5 and Figure 136–7). The method by which the PMD control function affects these variables is implementation dependent.

If the PMA is connected to the service interface of a PMD that includes the PMD control function and training is disabled by the management variable `mr_training_enable`, then `precoder_tx_out_enablei`, `precoder_rx_in_enablei`, `precoder_tx_in_enablei`, and `precoder_rx_out_enablei` are set as required by the implementation.

120.5.8 Link status

The PMA shall provide link status information to the PMA client using the `PMA:IS_SIGNAL.indication` primitive. The PMA continuously monitors the link status reported by the service interface below from the `inst:IS_SIGNAL.indication` primitive, and uses this as input to Signal Indication Logic (SIL) to determine the link status to report to the layer above. Other inputs to the SIL may include the status of clock and data recovery on the lanes from the service interface below the PMA and whether buffers/FIFOs have reached the required fill level to accommodate Skew Variation so that data is being sent on the output lanes.

120.5.9 PMA local loopback mode (optional)

PMA local loopback mode is optional. If it is implemented, it shall be as described in this subclause (120.5.9).

The PMA sublayer may provide a local loopback function. The function involves looping back each input lane to the corresponding output lane. Each bit received from the `PMA:IS_UNITDATAi.request(tx_symbol)` primitive is looped back in the direction of the MAC using the `PMA:IS_UNITDATAi.indication(rx_symbol)` primitive.

During local loopback, the PMA performs normal bit muxing of PCSs per 120.5.2 onto the lanes in the Tx direction toward the service interface below the PMA.

Ability to perform this function is indicated by the `Local_loopback_ability` status variable. If a Clause 45 MDIO is implemented, this variable is accessible through bit 1.8.0 (45.2.1.7.15). A device is placed in local loopback mode when the `Local_loopback_enable` control variable is set to one, and removed from local loopback mode when this variable is set to zero. If a Clause 45 MDIO is implemented, this variable is accessible through PMA/PMD control 1 register (bit 1.0.0, see 45.2.1.1.5).

120.5.10 PMA remote loopback mode (optional)

PMA remote loopback mode is optional. If implemented, it shall be as described in this subclause (120.5.10).

Remote loopback, if provided, should be implemented in a PMA sublayer close enough to the PMD to maintain the bit sequence on each individual PMD lane. When remote loopback is enabled, each symbol received over a lane of the service interface below the PMA via `inst:IS_UNITDATA_i.indication` is looped back to the corresponding output lane toward the PMD via `inst:IS_UNITDATA_i.request`. Note that the service interface below the PMA can be provided by the PHY XS, PMD, or another PMA sublayer.

During remote loopback, the PMA performs normal bit muxing of PCSs per 120.5.2 onto the lanes in the Rx direction towards the PMA client.

The ability to perform this function is indicated by the `200G_Remote_loopback_ability` and `400G_Remote_loopback_ability` status variables for the 200GBASE-R PMA and 400GBASE-R PMA, respectively. If a Clause 45 MDIO is implemented, the `200G_Remote_loopback_ability` and `400G_Remote_loopback_ability` variables are accessible through bit 1.23.15 (45.2.1.21.1) and bit 1.24.15 (45.2.1.22.1), respectively. A device is placed in remote loopback mode when the `Remote_loopback_enable` control variable is set to one, and removed from remote loopback mode when this variable is set to zero. If a Clause 45 MDIO is implemented, this variable is accessible through PMA/PMD Control register 1 (bit 1.0.1, see 45.2.1.1.4).

120.5.11 PMA test patterns (optional)

Where the output lanes of the PMA appear on a physically instantiated interface 200GAUI-n, 400GAUI-n or the PMD service interface (whether or not it is physically instantiated), the PMA may optionally generate and detect test patterns. These test patterns are used to test adjacent layer interfaces for an individual PMA sublayer or to perform testing between a physically instantiated interface of a PMA sublayer and external testing equipment.

120.5.11.1 Test patterns for NRZ encoded signals

For a 200GBASE-R PMA with 8 output lanes or a 400GBASE-R PMA with 16 output lanes using 2-level NRZ encoding, the test patterns in this clause may be supported.

The ability to generate each of the respective test patterns in each direction of transmission are indicated by the `PRBS9_Tx_generator_ability`, `PRBS9_Rx_generator_ability`, `PRBS31_Tx_generator_ability`, `PRBS31_Rx_generator_ability`, and `Square_wave_ability` status variables, which if a Clause 45 MDIO is implemented are accessible through bits 1.1500.5, 1.1500.4, 1.1500.3, 1.1500.1, and 1.1500.12, respectively (see 45.2.1.169).

120.5.11.1.1 PRBS31 test pattern

The ability to check PRBS31 test patterns in each direction of transmission are indicated by the `PRBS31_Tx_checker_ability` and `PRBS31_Rx_checker_ability` status variables, which, if a Clause 45 MDIO is implemented, are accessible through bits 1.1500.2 and 1.1500.0, respectively (see 45.2.1.169).

If supported, when send Tx PRBS31 test pattern is enabled by the PRBS31_enable and PRBS_Tx_gen_enable control variables, the PMA shall generate a PRBS31 pattern (as defined in 49.2.8) on each of the lanes toward the service interface below the PMA via the *inst:IS_UNITDATA_i.request* primitive. To avoid correlated crosstalk, it is highly recommended that the PRBS31 patterns generated on each lane be generated from independent, random seeds or at a minimum offset of 20 000 UI between the PRBS31 sequence on any lane and any other lane. When send Tx PRBS31 test pattern is disabled, the PMA returns to normal operation performing bit multiplexing as described in 120.5.2. If a Clause 45 MDIO is implemented, the PRBS31_enable and PRBS_Tx_gen_enable control variables are accessible through bits 1.1501.7 and 1.1501.3 (see 45.2.1.170).

If supported, when send Rx PRBS31 test pattern is enabled by the PRBS31_enable and PRBS_Rx_gen_enable control variables, the PMA shall generate a PRBS31 pattern on each of the lanes toward the PMA client via the PMA:IS_UNITDATA_i.indication primitive. While this test pattern is enabled, the PMA also generates PMA:IS_SIGNAL.indication(SIGNAL_OK) toward the PMA client independent of the link status at the service interface below the PMA. When send Rx PRBS31 test pattern is disabled, the PMA returns to normal operation performing bit multiplexing as described in 120.5.2. If a Clause 45 MDIO is implemented, the PRBS31_enable and PRBS_Rx_gen_enable control variables are accessible through bits 1.1501.7 and 1.1501.1 (see 45.2.1.170).

If supported, when check Tx PRBS31 test pattern mode is enabled by the PRBS31_enable and PRBS_Tx_check_enable control variables, the PMA shall check for the PRBS31 pattern on each of the lanes received from the PMA client via the PMA:IS_UNITDATA_i.request primitive. The checker shall increment the test-pattern error counter by one for each incoming bit error in the PRBS31 pattern for isolated single bit errors. Implementations should be capable of counting at least one error whenever one or more errors occur in a sliding 1000-bit window. If a Clause 45 MDIO is implemented, the PRBS31_enable and PRBS_Tx_check_enable control variables are accessible through bits 1.1501.7 and 1.1501.2 (see 45.2.1.170). The Tx test-pattern error counters Ln0_PRBS_Tx_test_err_counter through Ln15_PRBS_Tx_test_err_counter count, per lane, errors in detecting the PRBS31 pattern on the lanes from the PMA client. If a Clause 45 MDIO is implemented, these counters are accessible through registers 1.1600 through 1.16015 (see 45.2.1.173). When check Tx PRBS31 test pattern is disabled, the PMA expects normal traffic and test-pattern error counting does not continue. While in check Tx PRBS31 test-pattern mode, bit multiplexing continues as described in 120.5.2. Note that bit multiplexing of per-lane PRBS31 may produce a signal which is not meaningful for downstream sublayers.

If supported, when check Rx PRBS31 test-pattern mode is enabled by the PRBS31_enable and PRBS_Rx_check_enable control variables, the PMA shall check for the PRBS31 pattern on each of the lanes received from the service interface below the PMA via the *inst:IS_UNITDATA_i.indication* primitive. If a Clause 45 MDIO is implemented, the PRBS31_enable and PRBS_Rx_check_enable control variables are accessible through bits 1.1501.7 and 1.1501.0 (see 45.2.1.170). The Rx test-pattern error counters Ln0_PRBS_Rx_test_err_counter through Ln15_PRBS_Rx_test_err_counter count, per lane, errors in detecting the PRBS31 pattern on the lanes from the service interface below the PMA. If a Clause 45 MDIO is implemented, these counters are accessible through registers 1.1700 through 1.1715 (see 45.2.1.174). While in check Rx PRBS31 test-pattern mode, the PMA:IS_SIGNAL.indication primitive does not indicate a valid signal. When check Rx PRBS31 test pattern is disabled, the PMA returns to normal operation performing bit multiplexing as described in 120.5.2.

120.5.11.1.2 PRBS9 test pattern

If supported, when send Tx PRBS9 test-pattern mode is enabled by the PRBS9_enable and Tx_gen_enable control variables, the PMA shall generate a PRBS9 pattern (as defined in Table 68–6) on each lane toward the service interface below the PMA via the *inst:IS_UNITDATA_i.request* primitive. If a Clause 45 MDIO is implemented, the PRBS9_enable and Tx_gen_enable control variables are accessible through bits 1.1501.6 and 1.1501.3 (see 45.2.1.170). When send Tx PRBS9 test-pattern mode is disabled, the PMA returns to normal operation performing bit multiplexing as described in 120.5.2.

If supported, when send Rx PRBS9 test-pattern mode is enabled by the PRBS9_enable and Rx_gen_enable control variables, the PMA shall generate a PRBS9 pattern on each lane toward the PMA client via the PMA:IS_UNITDATA_i.indication primitive. The PMA also generates PMA:IS_SIGNAL.indication(SIGNAL_OK) toward the PMA client independent of the link status at the service interface below the PMA. If a Clause 45 MDIO is implemented, the PRBS9_enable and Rx_gen_enable control variables are accessible through bits 1.1501.6 and 1.1501.1 (see 45.2.1.170). When send Rx PRBS9 test-pattern mode is disabled, the PMA returns to normal operation performing bit multiplexing as described in 120.5.2.

Note that PRBS9 is intended to be checked by external test gear, and no PRBS9 checking function is provided within the PMA.

120.5.11.1.3 Square wave test pattern

Transmit square wave test-pattern mode optionally applies to each lane of the Tx direction PMA towards a physically instantiated 200GAUI-8, 400GAUI-16 or towards an NRZ PMD service interface whether or not it is physically instantiated. The ability to perform this function is indicated by the Square_wave_ability status variable. If a Clause 45 MDIO is implemented, the Square_wave_ability status variable is accessible through the Square wave test ability bit 1.1500.12 (see 45.2.1.169). If implemented, the transmit square wave test-pattern mode is enabled by control variables Square_wave_enable_0 through Square_wave_enable_15. If a Clause 45 MDIO is implemented, these control variables are accessible through the square wave testing control and status register bits 1.1510.0 through 1.1510.15 (limited to the number of lanes of the service interface below the PMA, see 45.2.1.171). When enabled, the PMA shall generate a square wave test pattern (8 ones followed by 8 zeros) on the square wave enabled lanes toward the service interface below the PMA via the *inst*:IS_UNITDATA_i.request primitive. Lanes for which square wave is not enabled transmit normal data resulting from the bit multiplexing operations described in 120.5.2 or test patterns as determined by other registers. When transmit square wave test pattern is disabled for all lanes, the PMA performs normal operation performing bit multiplexing as described in 120.5.2 or transmit test patterns as determined by other registers.

120.5.11.2 Test patterns for PAM4 encoded signals

For a 200GBASE-R PMA with 4 output lanes or a 400GBASE-R PMA with 4 or 8 output lanes using PAM4 encoding, the test patterns described in this clause may optionally be supported.

The patterns PRBS13Q and square wave (quaternary) can be enabled on a lane-by-lane basis. The patterns PRBS31Q and SSPRQ can be enabled on all lanes of an interface at once. If per-lane pattern(s) are enabled for a subset of the lanes and a per-interface pattern is also enabled, the per-lane patterns are generated only on the indicated lanes and the per-interface pattern is generated on the remaining lanes. The behavior if more than one per-lane pattern is enabled for the same lane or more than one per-interface pattern is enabled is not defined.

120.5.11.2.1 PRBS13Q test pattern

A PMA may optionally include a PRBS13Q pattern generator as specified in this subclause. The ability to perform this function is indicated by the PRBS13Q_gen_Tx_ability and PRBS13Q_gen_Rx_ability status variables, reflecting the ability to send this test pattern in the direction towards the PMD and towards the MAC, respectively. If a Clause 45 MDIO is implemented, the PRBS13Q_gen_Tx_ability and PRBS13Q_gen_Rx_ability status variables are accessible through the PRBS13Q Tx generator ability and PRBS13Q Rx generator ability bits 1.1500.11 and 1.1500.10 (see 45.2.1.169).

When the PRBS13Q test pattern enabled, it replaces the signal on the output lane(s) for which it is enabled. The PRBS13Q test pattern is a repeating 8191-symbol sequence formed by Gray coding pairs of bits from two repetitions of the PRBS13 pattern into PAM4 symbols as described in 120.5.7.1. The PRBS pattern

generator produces the same result as the implementation shown in Figure 94–6, which implements the generator polynomial shown in Equation (94–3). Since the PRBS13 pattern is an odd number of bits in length, bits which are mapped as the first bit of a PAM4 symbol during one repetition of the PRBS13 sequence are mapped as the second bit of a PAM4 symbol during the next repetition of the PRBS13 sequence, and bits which are mapped as the second bit of a PAM4 symbol are mapped as the first bit of the following symbol in the next repetition of the PRBS13 sequence. For example, if the PRBS13 generator used to create the PRBS13Q sequence is initialized to a seed value of 0000010101011 (with the leftmost bit in S0 and the rightmost in S12), the PRBS13Q sequence begins with the following Gray coded PAM4 symbols, transmitted left to right: 1031320220111130103121231210012102121023131112.

The PRBS13Q test pattern is enabled in the transmit direction by the Tx_gen_enable and PRBS13Q_enable_0 through PRBS13Q_enable_7 control variables (in the downstream direction towards the PMD), and in the receive direction by the Rx_gen_enable and PRBS13Q_enable_0 through PRBS13Q_enable_7 control variables (in the upstream direction towards the MAC). If the optional Clause 45 MDIO is implemented, the Tx_gen_enable and Rx_gen_enable control variables are accessible through the Tx generator enable and Rx generator enable bits 1.1501.3 and 1.1501.1 (see 45.2.1.170), and the PRBS13_enable_0 through PRBS13_enable_7 control variables are accessible through bits 1.1512.0 through 1.1512.7 (see 45.2.1.172).

120.5.11.2.2 PRBS31Q test pattern

A PMA may optionally include a PRBS31Q pattern generator as specified in this subclause. The ability to generate PRBS31Q patterns in each direction of transmission are indicated by the PRBS31Q_gen_Tx_ability and PRBS31Q_gen_Rx_ability status variables, reflecting the ability to send this test pattern in the direction towards the PMD and towards the MAC, respectively. The ability to check PRBS31Q patterns in each direction of transmission are indicated by the PRBS31Q_Tx_checker_ability and PRBS31Q_Rx_checker_ability status variables. If a Clause 45 MDIO is implemented, the PRBS31Q_gen_Tx_ability, PRBS31Q_gen_Rx_ability, PRBS31Q_Tx_checker_ability and PRBS31Q_Rx_checker_ability status variables are accessible through the PRBS31Q Tx generator ability, PRBS31Q Rx generator ability, PRBS31Q Tx checker ability, and PRBS31Q Rx checker ability bits 1.1500.9, 1.1500.7, 1.1500.8, and 1.1500.6 (see 45.2.1.169).

The PRBS31Q test pattern is a repeating $2^{31}-1$ -symbol sequence formed by Gray coding pairs of bits from two repetitions of the PRBS31 pattern defined in 49.2.8 into PAM4 symbols as described in 120.5.7.1. Since the PRBS31 pattern is an odd number of bits in length, bits that are mapped as the first bit of a PAM4 symbol during one repetition of the PRBS31 sequence are mapped as the second bit of a PAM4 symbol during the next repetition of the PRBS31 sequence, and bits that are mapped as the second bit of a PAM4 symbol during one repetition of the PRBS31 sequence are mapped as the first bit of the following symbol in the next repetition of the PRBS31 sequence. For example, if the PRBS31 generator used to create the PRBS31Q sequence is initialized to a seed value of all ones, the PRBS31Q sequence begins with the following Gray coded PAM4 symbols, transmitted left to right: 22222222222220122222222222000222222222201201222. To avoid correlated crosstalk, it is highly recommended that the PRBS31 patterns used to generate the PRBS31Q pattern on each lane are generated from independent, random seeds, or at a minimum offset of 20 000 UI between the PRBS31 sequence used to generate the PRBS31Q pattern on any lane and any other lane. A PRBS31Q pattern checker operates by converting PAM4 symbols received on each input lane to pairs of bits as described in 120.5.7.1 and then using a PRBS31 pattern checker on the resulting bit stream. The checker shall increment the test-pattern error counter by one for each incoming bit error in the PRBS31 pattern for isolated single bit errors. Implementations should be capable of counting at least one error whenever one or more errors occur in a sliding 1000-bit window.

If supported, when send Tx PRBS31Q test pattern is enabled by the PRBS31Q_pattern_enable and PRBS_Tx_gen_enable control variables, the PMA shall generate a PRBS31Q pattern on each of the lanes toward the service interface below the PMA via the *inst:IS_UNITDATA_i.request* primitive. When send Tx

PRBS31Q test pattern is disabled, the PMA returns to normal operation performing bit multiplexing as described in 120.5.2. If a Clause 45 MDIO is implemented, the PRBS31Q_pattern_enable and PRBS_Tx_gen_enable control variables are accessible through bits 1.1501.13 and 1.1501.3 (see 45.2.1.170).

If supported, when send Rx PRBS31Q test pattern is enabled by the PRBS31Q_pattern_enable and PRBS_Rx_gen_enable control variables, the PMA shall generate a PRBS31Q pattern on each of the lanes toward the PMA client via the PMA:IS_UNITDATA_i.indication primitive. The PMA also generates PMA:IS_SIGNAL.indication(SIGNAL_OK) toward the PMA client independent of the link status at the service interface below the PMA. If a Clause 45 MDIO is implemented, the PRBS31Q_pattern_enable and PRBS_Rx_gen_enable control variables are accessible through bits 1.1501.13 and 1.1501.1 (see 45.2.1.170). When send Rx PRBS31Q test pattern is disabled, the PMA returns to normal operation performing bit multiplexing as described in 120.5.2.

If supported, when check Tx PRBS31Q test pattern mode is enabled by the PRBS31Q_pattern_enable and PRBS_Tx_check_enable control variables, the PMA checks for the PRBS31Q pattern on each of the lanes received from the PMA client via the PMA:IS_UNITDATA_i.request primitive. If a Clause 45 MDIO is implemented, the PRBS31Q_pattern_enable and PRBS_Tx_check_enable control variables are accessible through bits 1.1501.13 and 1.1501.2 (see 45.2.1.170). The Tx test-pattern error counters Ln0_PRBS_Tx_test_err_counter through either Ln3_PRBS_Tx_test_error_counter or Ln7_PRBS_Tx_test_error_counter (depending on whether the number of lanes is 4 or 8) count, per lane, errors in detecting the PRBS31 pattern resulting from converting the PAM4 symbols received on each lane to pairs of bits. If a Clause 45 MDIO is implemented, these counters are accessible through registers 1.1600 through 1.1603 or 1.1607 (depending on whether the number of lanes is 4 or 8) (see 45.2.1.173). When check Tx PRBS31Q test pattern is disabled, the PMA expects normal traffic and test pattern error counting does not continue. While in check Tx PRBS31Q mode, bit multiplexing continues as described in 120.5.2. Note that bit multiplexing of per-lane PRBS31Q may produce a signal which is not meaningful for downstream sublayers.

If supported, when check Rx PRBS31Q test pattern mode is enabled by the PRBS31Q_pattern_enable and PRBS_Rx_check_enable control variables, the PMA checks for the PRBS31Q pattern on each of the lanes received from the service interface below the PMA via the *inst*:IS_UNITDATA_i.indication primitive. If a Clause 45 MDIO is implemented, the PRBS31Q_pattern_enable and PRBS_Rx_check_enable control variables are accessible through bits 1.1501.13 and 1.1501.0 (see 45.2.1.170). The Rx test-pattern error counters Ln0_PRBS_Rx_test_err_counter through either Ln3_PRBS_Rx_test_error_counter or Ln7_PRBS_Rx_test_error_counter (depending on whether the number of lanes is 4 or 8) count, per lane, errors in detecting the PRBS31 pattern resulting from converting the PAM4 symbols received on each lane to pairs of bits. If a Clause 45 MDIO is implemented, these counters are accessible through registers 1.1700 through 1.1703 or 1.1707 (depending on whether the number of lanes is 4 or 8) (see 45.2.1.174). While in check Rx PRBS31Q mode, the PMA:IS_SIGNAL.indication primitive does not indicate a valid signal. When check Rx PRBS31Q test pattern is disabled, the PMA returns to normal operation performing bit multiplexing as described in 120.5.2.

120.5.11.2.3 SSPRQ test pattern

A PMA may optionally include a short stress pattern random quaternary (SSPRQ) test pattern generator as specified in this subclause. The ability to perform this function is indicated by the SSPRQ_gen_Tx_ability status variable, reflecting the ability to send this test pattern in the direction towards the PMD. If a Clause 45 MDIO is implemented, the SSPRQ_gen_Tx_ability status variable is accessible through the SSPRQ Tx generator ability bit 1.1500.13 (see 45.2.1.169).

The SSPRQ pattern is a repeating $2^{16}-1$ PAM4 symbol sequence constructed as follows:

- Bit sequence A is a 32768-bit sequence composed of three sections of the PRBS31 binary sequence (see 120.5.11.1.1) according to Table 120–2.

Table 120–2—SSPRQ bit sequence A

Pattern	Seed	Length
PRBS31	0x00000002	10924 bits
	0x34013FF7	10922 bits
	0x0CCCCCCC	10922 bits

- Each section of PRBS31 is generated as if produced by the shift register implementation shown in Figure 49–9 and the seed is a 31-bit hexadecimal value used to preset S30 through S0 (S30 is set to the MSB and S0 is set to the LSB) prior to the generation of the PRBS31 sequence for the indicated length of bits.
- Bit sequence B is a 65534-bit sequence formed by removing the first and last binary bit from a sequence consisting of two repetitions of bit sequence A.
- PAM4 sequence 1 is a 16384-symbol sequence formed by Gray coding bit sequence A according to 120.5.7.1.
- PAM4 sequence 2 is a 16384-symbol sequence formed by Gray coding bit sequence A according to 120.5.7.1, and then inverting each PAM4 symbol by replacing each PAM4 symbol n with the value $3-n$.
- PAM4 sequence 3 is a 16383-symbol sequence formed by Gray coding the first 32766 bits of bit-sequence B according to 120.5.7.1.
- PAM4 sequence 4 is a 16384-symbol sequence formed by Gray coding the last 32768 bits of bit-sequence B according to 120.5.7.1, and then inverting each PAM4 symbol by replacing each PAM4 symbol n with the value $3-n$.
- The repeating SSPRQ pattern formed by concatenating PAM4 sequences 1, 2, 3 and 4.

NOTE—A CSV file containing the entire PAM4 symbol sequence for the SSPRQ test pattern is available at <https://standards.ieee.org/downloads/802.3/>.

If supported, when send SSPRQ test pattern is enabled by the SSPRQ_pattern_enable control variable, the PMA shall generate an SSPRQ pattern on each of its lanes in the Tx direction towards the PMD with at least a 31 UI delay between the SSPRQ pattern on one lane and any other lane. If a Clause 45 MDIO is implemented, the SSPRQ_pattern_enable control variable is accessible through the SSPRQ pattern enable bit 1.1501.14 (see 45.2.1.170).

120.5.11.2.4 Square wave (quaternary) test pattern

Transmit square wave (quaternary) test-pattern mode optionally applies to each lane of the Tx direction PMA towards a physically instantiated 200GAUI-4, 400GAUI-8 or towards a PAM4 PMD service interface whether or not it is physically instantiated.

The ability to perform this function is indicated by the Square_wave_ability status variable. If a Clause 45 MDIO is implemented, the Square_wave_ability status variable is accessible through the Square wave test ability bit 1.1500.12 (see 45.2.1.169). If implemented, the transmit square wave test-pattern mode is enabled by control variables Square_wave_enable_0 through Square_wave_enable_8. If a Clause 45 MDIO is implemented, these control variables are accessible through the square wave testing control and status register bits 1.1510.0 through 1.1510.15 (limited to the number of lanes of the service interface below the PMA, see 45.2.1.171). When enabled, the PMA shall generate a square wave test pattern (8 threes followed

by 8 zeros) on the square wave enabled lanes toward the service interface below the PMA via the *inst:IS_UNITDATA_i.request* primitive. Lanes for which square wave is not enabled transmit normal data resulting from the bit multiplexing operations described in 120.5.2 or test patterns as determined by other registers. When transmit square wave test pattern is disabled for all lanes, the PMA performs normal operation performing bit multiplexing as described in 120.5.2 or transmit test patterns as determined by other registers.

NOTE—A square wave transmitted over a 200GAUI-4 or 400GAUI-8 may not be correctly forwarded to the PMD transmitter output.

120.6 PMA MDIO function mapping

The optional MDIO capability described in Clause 45 describes several variables that provide control and status information for and about the PMA. Since a given implementation may employ more than one PMA sublayer, the PMA control and status information is organized into multiple addressable instances, one for each possible PMA sublayer. See 45.2.1 and 120.1.4 for the allocation of MMD addresses to PMA sublayers. Control and status registers for MMD 8, 9, and 10 use the Extended PMA control and status registers at identical locations to those for MMD 1.

Mapping of MDIO control variables to PMA control variables is shown in Table 120–3. Mapping of MDIO status variables to PMA status variables is shown in Table 120–4. Mapping of MDIO counter to PMA counters is shown in Table 120–5. These tables provide the register and bit numbers for the PMA addressed as MMD 1. For implementations with multiple PMA sublayers, additional PMA sublayers use the corresponding register and bit numbers in MMDs 8, 9, and 10 as necessary.

Table 120–3—MDIO/PMA control variable mapping

MDIO variable	PMA/PMD register name	Register/ bit number	PMA control variable
PMA remote loopback	PMA/PMD control 1	1.0.1	Remote_loopback_enable
PMA local loopback	PMA/PMD control 1	1.0.0	Local_loopback_enable
Lane 0 Tx output precoder enable	PMA precoder control Tx output	1.600.0	precoder_tx_out_enable_0
Lane 1 Tx output precoder enable	PMA precoder control Tx output	1.600.1	precoder_tx_out_enable_1
Lane 2 Tx output precoder enable	PMA precoder control Tx output	1.600.2	precoder_tx_out_enable_2
Lane 3 Tx output precoder enable	PMA precoder control Tx output	1.600.3	precoder_tx_out_enable_3
Lane 0 Rx input precoder enable	PMA precoder control Rx input	1.601.0	precoder_rx_in_enable_0
Lane 1 Rx input precoder enable	PMA precoder control Rx input	1.601.1	precoder_rx_in_enable_1
Lane 2 Rx input precoder enable	PMA precoder control Rx input	1.601.2	precoder_rx_in_enable_2
Lane 3 Rx input precoder enable	PMA precoder control Rx input	1.601.3	precoder_rx_in_enable_3
PRBS31 pattern enable	PRBS pattern testing control	1.1501.7	PRBS31_enable
PRBS9 pattern enable	PRBS pattern testing control	1.1501.6	PRBS9_enable
PRBS31Q pattern enable	PRBS pattern testing control	1.1501.13	PRBS31Q_pattern_enable
SSPRQ pattern enable	PRBS pattern testing control	1.1501.14	SSPRQ_pattern_enable
Tx generator enable	PRBS pattern testing control	1.1501.3	PRBS_Tx_gen_enable

Table 120–3—MDIO/PMA control variable mapping (*continued*)

MDIO variable	PMA/PMD register name	Register/ bit number	PMA control variable
Tx checker enable	PRBS pattern testing control	1.1501.2	PRBS_Tx_check_enable
Rx generator enable	PRBS pattern testing control	1.1501.1	PRBS_Rx_gen_enable
Rx checker enable	PRBS pattern testing control	1.1501.0	PRBS_Rx_check_enable
Lane 0 SW enable	Square wave testing control	1.1510.0	Square_wave_enable_0
Lane 1 SW enable	Square wave testing control	1.1510.1	Square_wave_enable_1
Lane 2 SW enable	Square wave testing control	1.1510.2	Square_wave_enable_2
Lane 3 SW enable	Square wave testing control	1.1510.3	Square_wave_enable_3
Lane 4 SW enable	Square wave testing control	1.1510.4	Square_wave_enable_4
Lane 5 SW enable	Square wave testing control	1.1510.5	Square_wave_enable_5
Lane 6 SW enable	Square wave testing control	1.1510.6	Square_wave_enable_6
Lane 7 SW enable	Square wave testing control	1.1510.7	Square_wave_enable_7
Lane 8 SW enable	Square wave testing control	1.1510.8	Square_wave_enable_8
Lane 9 SW enable	Square wave testing control	1.1510.9	Square_wave_enable_9
Lane 10 SW enable	Square wave testing control	1.1510.10	Square_wave_enable_10
Lane 11 SW enable	Square wave testing control	1.1510.11	Square_wave_enable_11
Lane 12 SW enable	Square wave testing control	1.1510.12	Square_wave_enable_12
Lane 13 SW enable	Square wave testing control	1.1510.13	Square_wave_enable_13
Lane 14 SW enable	Square wave testing control	1.1510.14	Square_wave_enable_14
Lane 15 SW enable	Square wave testing control	1.1510.15	Square_wave_enable_15
Lane 0 PRBS13Q enable	PRBS13Q testing control	1.1512.0	PRBS13Q_enable_0
Lane 1 PRBS13Q enable	PRBS13Q testing control	1.1512.1	PRBS13Q_enable_1
Lane 2 PRBS13Q enable	PRBS13Q testing control	1.1512.2	PRBS13Q_enable_2
Lane 3 PRBS13Q enable	PRBS13Q testing control	1.1512.3	PRBS13Q_enable_3
Lane 4 PRBS13Q enable	PRBS13Q testing control	1.1512.4	PRBS13Q_enable_4
Lane 5 PRBS13Q enable	PRBS13Q testing control	1.1512.5	PRBS13Q_enable_5
Lane 6 PRBS13Q enable	PRBS13Q testing control	1.1512.6	PRBS13Q_enable_6
Lane 7 PRBS13Q enable	PRBS13Q testing control	1.1512.7	PRBS13Q_enable_7

Table 120–4—MDIO/PMA status variable mapping

MDIO status variable	PMA/PMD register name	Register/bit number	PMA status variable
PMA remote loopback ability	200G PMA/PMD extended ability register	1.23.15	200G_Remote_loopback_ability
PMA remote loopback ability	400G PMA/PMD extended ability register	1.24.15	400G_Remote_loopback_ability
PMA local loopback ability	PMA/PMD status 2 register	1.8.0	Local_loopback_ability
PRBS9 Tx generator ability	Test-pattern ability register	1.1500.5	PRBS9_Tx_generator_ability
PRBS9 Rx generator ability	Test-pattern ability register	1.1500.4	PRBS9_Rx_generator_ability
PRBS31Tx generator ability	Test-pattern ability register	1.1500.3	PRBS31_Tx_generator_ability
PRBS31Tx checker ability	Test-pattern ability register	1.1500.2	PRBS31_Tx_checker_ability
PRBS31 Rx generator ability	Test-pattern ability register	1.1500.1	PRBS31_Rx_generator_ability
PRBS31 Rx checker ability	Test-pattern ability register	1.1500.0	PRBS31_Rx_checker_ability
Square wave test ability	Test-pattern ability register	1.1500.12	Square_wave_ability
PRBS13Q Tx generator ability	Test-pattern ability register	1.1500.11	PRBS13Q_gen_Tx_ability
PRBS13Q Rx generator ability	Test-pattern ability register	1.1500.10	PRBS13Q_gen_Rx_ability
PRBS31Q Tx generator ability	Test-pattern ability register	1.1500.9	PRBS31Q_gen_Tx_ability
PRBS31Q Rx generator ability	Test-pattern ability register	1.1500.7	PRBS31Q_gen_rx_ability
PRBS31Q Tx checker ability	Test-pattern ability register	1.1500.8	PRBS31Q_Tx_checker_ability
PRBS31Q Rx checker ability	Test-pattern ability register	1.1500.6	PRBS31Q_Rx_checker_ability
SSPRQ Tx generator ability	Test-pattern ability register	1.1500.13	SSPRQ_gen_Tx_ability

Table 120–5—MDIO/PMA counters mapping

MDIO variable	PMA/PMD register name	Register/bit number	PMA status variable
Error counter Tx, lane 0	PRBS Tx pattern testing error counter, lane 0	1.1600	Ln0_PRBS_Tx_test_err_counter
Error counter Tx, lane 1	PRBS Tx pattern testing error counter, lane 1	1.1601	Ln1_PRBS_Tx_test_err_counter
Error counter Tx, lane 2	PRBS Tx pattern testing error counter, lane 2	1.1602	Ln2_PRBS_Tx_test_err_counter
Error counter Tx, lane 3	PRBS Tx pattern testing error counter, lane 3	1.1603	Ln3_PRBS_Tx_test_err_counter
Error counter Tx, lane 4	PRBS Tx pattern testing error counter, lane 4	1.1604	Ln4_PRBS_Tx_test_err_counter

Table 120–5—MDIO/PMA counters mapping (*continued*)

MDIO variable	PMA/PMD register name	Register/bit number	PMA status variable
Error counter Tx, lane 5	PRBS Tx pattern testing error counter, lane 5	1.1605	Ln5_PRBS_Tx_test_err_counter
Error counter Tx, lane 6	PRBS Tx pattern testing error counter, lane 6	1.1606	Ln6_PRBS_Tx_test_err_counter
Error counter Tx, lane 7	PRBS Tx pattern testing error counter, lane 7	1.1607	Ln7_PRBS_Tx_test_err_counter
Error counter Tx, lane 8	PRBS Tx pattern testing error counter, lane 8	1.1608	Ln8_PRBS_Tx_test_err_counter
Error counter Tx, lane 9	PRBS Tx pattern testing error counter, lane 9	1.1609	Ln9_PRBS_Tx_test_err_counter
Error counter Tx, lane 10	PRBS Tx pattern testing error counter, lane 10	1.1610	Ln10_PRBS_Tx_test_err_counter
Error counter Tx, lane 11	PRBS Tx pattern testing error counter, lane 11	1.1611	Ln11_PRBS_Tx_test_err_counter
Error counter Tx, lane 12	PRBS Tx pattern testing error counter, lane 12	1.1612	Ln12_PRBS_Tx_test_err_counter
Error counter Tx lane 13	PRBS Tx pattern testing error counter, lane 13	1.1613	Ln13_PRBS_Tx_test_err_counter
Error counter Tx lane 14	PRBS Tx pattern testing error counter, lane 14	1.1614	Ln14_PRBS_Tx_test_err_counter
Error counter Tx lane 15	PRBS Tx pattern testing error counter, lane 15	1.1615	Ln15_PRBS_Tx_test_err_counter
Error counter Rx, lane 0	PRBS Rx pattern testing error counter, lane 0	1.1700	Ln0_PRBS_Rx_test_err_counter
Error counter Rx, lane 1	PRBS Rx pattern testing error counter, lane 1	1.1701	Ln1_PRBS_Rx_test_err_counter
Error counter Rx, lane 2	PRBS Rx pattern testing error counter, lane 2	1.1702	Ln2_PRBS_Rx_test_err_counter
Error counter Rx, lane 3	PRBS Rx pattern testing error counter, lane 3	1.1703	Ln3_PRBS_Rx_test_err_counter
Error counter Rx, lane 4	PRBS Rx pattern testing error counter, lane 4	1.1704	Ln4_PRBS_Rx_test_err_counter
Error counter Rx, lane 5	PRBS Rx pattern testing error counter, lane 5	1.1705	Ln5_PRBS_Rx_test_err_counter
Error counter Rx, lane 6	PRBS Rx pattern testing error counter, lane 6	1.1706	Ln6_PRBS_Rx_test_err_counter
Error counter Rx, lane 7	PRBS Rx pattern testing error counter, lane 7	1.1707	Ln7_PRBS_Rx_test_err_counter
Error counter Rx, lane 8	PRBS Rx pattern testing error counter, lane 8	1.1708	Ln8_PRBS_Rx_test_err_counter
Error counter Rx, lane 9	PRBS Rx pattern testing error counter, lane 9	1.1709	Ln9_PRBS_Rx_test_err_counter

Table 120–5—MDIO/PMA counters mapping (*continued*)

MDIO variable	PMA/PMD register name	Register/bit number	PMA status variable
Error counter Rx, lane 10	PRBS Rx pattern testing error counter, lane 10	1.1710	Ln10_PRBS_Rx_test_err_counter
Error counter Rx, lane 11	PRBS Rx pattern testing error counter, lane 11	1.1711	Ln11_PRBS_Rx_test_err_counter
Error counter Rx, lane 12	PRBS Rx pattern testing error counter, lane 12	1.1712	Ln12_PRBS_Rx_test_err_counter
Error counter Rx, lane 13	PRBS Rx pattern testing error counter, lane 13	1.1713	Ln13_PRBS_Rx_test_err_counter
Error counter Rx, lane 14	PRBS Rx pattern testing error counter, lane 14	1.1714	Ln14_PRBS_Rx_test_err_counter
Error counter Rx, lane 15	PRBS Rx pattern testing error counter, lane 15	1.1715	Ln15_PRBS_Rx_test_err_counter

120.7 Protocol implementation conformance statement (PICS) proforma for Clause 120, Physical Medium Attachment (PMA) sublayer, type 200GBASE-R and 400GBASE-R²⁰⁷

120.7.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 120, Physical Medium Attachment (PMA) sublayer, type 200GBASE-R and 400GBASE-R, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

120.7.2 Identification

120.7.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

120.7.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3-2022, Clause 120, Physical Medium Attachment (PMA) sublayer, type 200GBASE-R and 400GBASE-R
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No ☐ Yes ☐ (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3-2022.)	

Date of Statement	
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²⁰⁷ *Copyright release for PICS proformas:* Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

120.7.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
*PMA200	PMA for 200GBASE-R	120.1.1		O.1	Yes [] No []
*PMA400	PMA for 400GBASE-R	120.1.1		O.1	Yes [] No []
*LNS_UP-STRM	Number of lanes in direction of MAC	120.1.4	Divisor of number of PCS lanes	PMA200: M	8 [] 4 []
				PMA400:M	16 [] 8 []
UP_NRZ	NRZ modulation used for PMA service interface	120.1.4		(PMA200 LNS_UPSTRM=8 or PMA400* LNS_UPSTRM=16):M	Yes [] N/A []
UP_PAM4	PAM4 modulation used for PMA service interface	120.1.4		(PMA200 LNS_UPSTRM=4 or PMA400* LNS_UPSTRM=8):M	Yes [] N/A []
*LNS_DN-STRM	Number of lanes in the direction of PMD	120.1.4	Divisor of number of PCS lanes	PMA200: M	8 [] 4 []
				PMA400:M	16 [] 8 [] 4 []
DN_NRZ	NRZ modulation used for service interface below the PMA	120.1.4		(PMA200 LNS_DNSTRM=8 or PMA400* LNS_DNSTRM=16):M	Yes [] N/A []
DN_PAM4	PAM4 modulation used for service interface below the PMA	120.1.4		(PMA200 LNS_DNSTRM=4 or PMA400* LNS_DNSTRM<16):M	Yes [] N/A []
RX_CLOCK	Signaling rate of output lanes in Rx direction	120.5.5	26.5625 GBd	M	Yes []
TX_CLOCK	Signaling rate of output lanes in Tx direction	120.5.5	26.5625 GBd	PMA200:M	Yes [] N/A []
			LNS_DNSTRM= 8 or 16: 26.5625 GBd LNS_DNSTRM= 4: 53.125 GBd	PMA400:M	Yes [] N/A []
LANE_MAPPING	Maintain lane mapping while link is in operation	120.5.2	Maintain sequence of PCSs on all output lanes	M	Yes []

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Item	Feature	Subclause	Value/Comment	Status	Support
LNKS	PMA link status	120.5.8	Meets the requirements of 120.5.8	M	Yes []
*LBL	PMA local loopback	120.5.9	Supports local loopback	O	Yes [] No []
*LBR	PMA remote loopback	120.5.10	Supports remote loopback	O	Yes [] No []
MD	MDIO	120.6	Registers and interface supported	O	Yes [] No []
*USP1SP6	Physically instantiated 200GAUI-n or 400GAUI-n above (toward MAC)	120.5.3		O	Yes [] No []
*DSP1SP6	Physically instantiated 200GAUI-n or 400GAUI-n below (toward PMD)	120.5.3		O	Yes [] No []
*SP2SP5	Physically instantiated PMD service interface	120.5.3		O	Yes [] No []
UNAU1	Electrical and timing requirements of Annex 120B, Annex 120C, Annex 120D, or Annex 120E as appropriate met by 200GAUI-n or 400GAUI-n of the PMA service interface	120.5.1, 120.5.5		USP1SP6:M	Yes [] N/A []
DNAUI	Electrical and timing requirements of Annex 120B, Annex 120C, Annex 120D, or Annex 120E as appropriate met by 200GAUI-n or 400GAUI-n of the service interface below the PMA	120.5.1, 120.5.5		DSP1SP6:M	Yes [] N/A []
DELAY200	Roundtrip delay limit for 200GBASE-R	120.5.4	No more than 18 432 BT or 36 pause_quanta	PMA200:M	Yes [] N/A []
DELAY400	Roundtrip delay limit for 400GBASE-R	120.5.4	No more than 36 864 BT or 72 pause_quanta	PMA400:M	Yes [] N/A []

120.7.4 Skew generation and tolerance

Item	Feature	Subclause	Value/Comment	Status	Support
S1	Skew generation toward SP1 in Tx direction	120.5.3.1	≤ 29 ns	DSP1S P6:M	Yes [] N/A []
S2	Skew variation generation toward SP1 in Tx direction	120.5.3.1	≤ 200 ps	DSP1S P6:M	Yes [] N/A []
S3	Skew variation tolerance at SP1	120.5.3.2	Minimum 200 ps	USP1S P6:M	Yes [] N/A []
S4	Skew generation toward SP2 in Tx direction	120.5.3.3	≤ 43 ns	SP2SP 5:M	Yes [] N/A []
S5	Skew variation generation toward SP2 in Tx direction	120.5.3.3	≤ 400 ps	SP2SP 5:M	Yes [] N/A []
S6	Skew variation tolerance at SP5	120.5.3.4	Minimum 3.6 ns	SP2SP 5:M	Yes [] N/A []
S7	Skew generation toward SP6 in Rx direction	120.5.3.5	≤ 160 ns	USP1S P6:M	Yes [] N/A []
S8	Skew variation generation toward SP6 in Rx direction	120.5.3.5	≤ 3.8 ns	USP1S P6:M	Yes [] N/A []
S9	Skew variation tolerance at SP6	120.5.3.6	Minimum 3.8 ns	DSP1S P6:M	Yes [] N/A []

120.7.5 Test patterns

Item	Feature	Subclause	Value/Comment	Status	Support
*JTP1	Physically Instantiated 200GAUI-n or 400GAUI-n between PMA and PMA client	120.5.11		O	Yes [] No []
*JTP2	Physically Instantiated 200GAUI-n or 400GAUI-n between PMA and sublayer below the PMA, or adjacent to PMD	120.5.11		O	Yes [] No []
J1	Send PRBS31 Tx	120.5.11.1.1		JTP2* DN_NRZ:O	Yes [] No [] N/A []
J2	Send PRBS31 Rx	120.5.11.1.1		JTP1* UP_NRZ:O	Yes [] No [] N/A []
J3	Check PRBS31 Tx	120.5.11.1.1		JTP1* UP_NRZ:O	Yes [] No [] N/A []
J4	Check PRBS31 Rx	120.5.11.1.1		JTP2* DN_NRZ:O	Yes [] No [] N/A []

Item	Feature	Subclause	Value/Comment	Status	Support
J5	Send PRBS9 Tx	120.5.11.1.2		JTP2* DN_NRZ:O	Yes [☐ No [☐ N/A [☐
J6	Send PRBS9 Rx	120.5.11.1.2		JTP1* UP_NRZ:O	Yes [☐ No [☐ N/A [☐
J7	Send square wave Tx	120.5.11.1.3		JTP2* DN_NRZ:O	Yes [☐ No [☐ N/A [☐
J8	Send PRBS13Q Tx	120.5.11.2.1		JTP2* DN_PAM4:O	Yes [☐ No [☐ N/A [☐
J9	Send PRBS13Q Rx	120.5.11.2.1		JTP1* UP_PAM4:O	Yes [☐ No [☐ N/A [☐
J10	Send PRBS31Q Tx	120.5.11.2.2		JTP2* DN_PAM4:O	Yes [☐ No [☐ N/A [☐
J11	Send PRBS31Q Rx	120.5.11.2.2		JTP1* UP_PAM4:O	Yes [☐ No [☐ N/A [☐
J12	Check PRBS31Q Tx	120.5.11.2.2		JTP1* UP_PAM4:O	Yes [☐ No [☐ N/A [☐
J13	Check PRBS31Q Rx	120.5.11.2.2		JTP2* DN_PAM4:O	Yes [☐ No [☐ N/A [☐
J14	Send SSPRQ Tx	120.5.11.2.3		DN_PAM4:O	Yes [☐ No [☐ N/A [☐
J15	Send square wave (quaternary) Tx	120.5.11.2.4		DN_PAM4:O	Yes [☐ No [☐ N/A [☐

120.7.6 Loopback modes

Item	Feature	Subclause	Value/Comment	Status	Support
LB1	PMA local loopback implemented	120.5.9	Meets the requirements of 120.5.9	LBL:M	Yes [☐ N/A [☐
LB2	PMA remote loopback implemented	120.5.10	Meets the requirements of 120.5.10	LBR:M	Yes [☐ N/A [☐

120.7.7 Encoding

Item	Feature	Subclause	Value/Comment	Status	Support
*CKR4	PMA is adjacent to a 200GBASE-CR4 or 200GBASE-KR4 PMD.	120.5.7.2		O	Yes [] No []
E1	PMA supports Tx precoding	120.5.7.2		CKR4:M	Yes [] N/A []
E2	PMA supports Rx precoding	120.5.7.2		CKR4:O	Yes [] No [] N/A []